MILITARY SPECIFICATION SHEET

ELECTRON TUBE, NEGATIVE GRID (MICROWAVE)

TYPE 8727

This specification is approved for use by Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, pencil, ceramic-metal

See figure 1

Mounting position: Any

Weight: 0.3 ounce (8.5 grams) nominal

ABSOLUTE RATINGS: Class C amplifier

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ik	Ic	Pp	TE	Barometric pressure, reduced
Unit:	V	Vdc	Vdc	V	Ohms	Meg	mAdc	mAdc	W	°C	mmHg
Maximum:	6.9	250	0	+50	---	0.25	25	6	2.5	225	8
Minimum:	5.7	---	-50	-50	---	---	---	---	---	---	---
TEST CONDITIONS:	6.3	125	Note 8	---	50	---	---	---	---	---	---

GENERAL:

Qualification - Not required

(B) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 1								
1266	Total grid current	3	Eb = 200 Vdc; Ec = -2 Vdc; Rg = 0.5 Meg; Rk = 0	0.65	II	Ic	0	-0.3	uAdc
1256	Electrode current (1) (anode)	-	Ck = 1,000 uF	0.65	II	Ib	9	19	mAdc
1256	Electrode current (2) (anode)	-	Ec = -5 Vdc; Rk = 0	0.65	II	Ib	---	50	uAdc
1306	Transconductance (1)	-	Ck = 1,000 uF	0.65	II	Sm	12,500	19,500	umhos
4250	Output power (1)	4	Eb = 250 Vdc	0.65	II	Po	1.7	---	W
	Quality conformance inspection, part 2								
1301	Heater current	-	Ef = 6.3 V	---	---	If	205	245	mA
1336	Heater-cathode leakage	-	Ehk = +60 Vdc Ehk = -60 Vdc	--- ---	--- ---	Ihk Ihk	--- ---	+30 -30	uAdc uAdc
1306	Transconductance (2)	5	Ef = 5.7 V	---	---	$\frac{\Delta Sm}{Ef}$	---	15	%
1331	Direct-interelectrode capacitance	1		---	---	Cgp Cgk Cpk	1.5 3.6 ---	2.7 5.00 0.040	pF pF pF
1256	Electrode current (3) (anode)	-	Ec = -2.5 Vdc; Rk = 0	---	---	Ib	100	---	uAdc
1316	Amplification factor	-	Ck = 1,000 uF	---	---	Mu	55	85	---
1211	Insulation of electrodes	-	E(g-all) = -100 Vdc E(p-all) = 300 Vdc	---	---	R	100	---	Meg
4250	Output power (2)	4	Ef = 5.7 V; Eb = 250 Vdc; do not retune cavity	---	---	$\frac{\Delta Po}{Ef}$	---	0.2	W
	Quality conformance inspection, part 3								
1031	Sweep frequency vibration	2,18	Ebb = 125 Vdc; R1 = 10,000 ohms; Rk = 50 ohms	---	---	Ep	---	150	mVac
1031	Variable-frequency vibration	2,18	Ebb = 125 Vdc; R1 = 10,000 ohms; Rk = 50 ohms	---	---	Ep	---	150	mVac
1041	Shock	16,18	Hammer angle = 30°; no voltages applied	---	---	---	---	---	---
1031	Vibration fatigue	17,18	2.5 G; fixed frequency; F = 25 Hz (min), 60 Hz (max)	---	---	---	---	---	---
---	Shock and vibration-fatigue test end points:								
1031	Sweep and variable-frequency vibration	-		---	---	Ep	---	200	mVac
1336	Heater-cathode leakage	-		---	---	Ihk	---	±60	uAdc
---	Seal fracture	6,7, 18		---	---	---	30	---	Lbs

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CLASS	SYMBOL	LIMITS MIN	LIMITS MAX	UNIT
	Quality conformance inspection, part 3 -Continued								
---	Seal strain	7,18, 19		---	---	---	---	---	---
(9) 1506	Heater-cycling life	9,12, 16	Group D; Ef = 6.3 V; Ehk = 0; Eb = 0; Rk = 0; Ec = 0	---	---	---	---	---	---
---	Heater-cycling life-test end points:								
1336	Heater-cathode leakage	-	Ehk = +60 Vdc	---	---	Ihk	---	+30	µAdc
			Ehk = -60 Vdc	---	---	Ihk	---	-30	µAdc
1211	Insulation of electrodes	-	E(g-all) = -100 Vdc	---	---	R	50	---	Meg
1516	Stability life test (2 hours)	11,12	Group D; Eb = 215 Vdc; Rk = 150 ohms; TA = room, 25°±5°C; g1 grounded	---	---	---	---	---	---
---	Stability life-test end point:								
1306	Transconductance (1)	-		---	---	ΔSm t	---	15	%
1521	Survival-rate life	13,15	Stability life test, or equivalent conditions; TA = room, 25° ±5°C; t = 100 hours	---	---	---	---	---	---
---	Survival-rate life test end points:								
1256	Electrode current (2) (anode)	-		---	---	Ib	---	50	µAdc
1306	Transconductance (1)	-		---	---	Sm	9,000	---	µmhos
1501	Intermittent life	10,14	Group C; Ef = 6.3 V; Ebb = 250 Vdc	---	---	t	2,000	---	cycles
---	Intermittent life-test end points:								
4250	Output power (1)	-		---	---	Po	0.90	---	W
1266	Total grid current	-		---	---	Ic	---	-1.0	µAdc
1211	Insulation of electrodes	-		---	---	R	60	---	Meg
1002	Barometric pressure, reduced	18	Pressure = 8 ±2 mmHg; voltage = 300 Vac	---	---	---	---	---	---

NOTES:

1. Test in capacitance jig RCA Capacity Adaptor Model H-1756E, or equivalent. A capacitance bridge operating frequency of 1.0 to 2.0 kHz may be used.
2. The tube shall be vibrated perpendicular to the major tube axis through a frequency range from 5 to 500 Hz and back. From 5 to 50 Hz, the tube shall be vibrated at a constant displacement of 0.040 ±0.0025 inch (1.02 ±0.06 mm). From 50 to 500 Hz, the tube shall be vibrated at a constant acceleration of 10 ±2G. Total time to complete a sweep cycle shall be 10 ±5 minutes.
   a. Each tube shall be vibrated for 60 seconds at the frequency which gives maximum vibrational output. If at the end of 60 seconds the vibrational output is still increasing, the vibration shall continue until there is no further increase.

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NOTES: -Continued

2. -Continued

   b. The tube shall be rejected for vibrational output voltage exceeding the limit specified herein.

3. This test is to be the first test performed at the conclusion of the holding period.

4. This test shall be performed in amplifier cavity Amerac Model No. 1244-1, or equivalent. Test frequency shall be 550 ±10 MHz. Input = 0.2 W. When the amplifier has been tuned, adjust Ec for Ib = 20 mAdc.

5. Transconductance (2) is the percent change in transconductance (1) of an individual tube resulting from a change in Ef.

6. Tube supported at cathode and anode terminals with the grid disk centered. The supports shall be placed 0.938 ±0.016 inch (23.83 ±0.41 mm) apart. Apply force to the grid disk without shock, perpendicular to the axis of the tube. Inspect for defects described in MIL-E-1, appendix D, paragraph Vacuum seal (metal-to-glass) defect classification.

7. Electrical rejects may be used.

(8) 8. Whenever Ec is not specified on any test, the grid shall be grounded directly or through a grid resistor (if specified) except when related MIL-STD-1311 test methods imply grid variance to accomplish this test.

9. The no-load to steady-state full load of the heater voltage supply shall be not more than 3 percent. A failure or defect shall consist of an open heater, open cathode circuit, heater cathode short, or heater cathode leakage in excess of the specified heater cycling life-test end-point limit.

10. Test shall be performed in amplifier cavity Amerac Model No. 1244-1, or equivalent. Test frequency shall be 550 ±10 MHz. Adjust Rk for Ik = 25 mAdc and Rg for Ig = 6 mAdc.

11. Cycle Ef only, 110 minutes "on", 10 minutes "off".

12. Test shall be performed on a group D basis every 90 days with the lot consisting of the current month and two subsequent months. Failure of any tube shall revert back to group D monthly lot sampling. After three consecutive successful lot submissions have been completed, the test may revert to the quarterly basis.

13. Test shall be performed in accordance with method 1521 and MIL-E-1 every 90 days. Failure of any tube shall revert back to MIL-E-1 monthly lot sampling. After three consecutive successful submissions have been completed, the test shall revert to the quarterly basis.

14. A complete cycle is made in a 6-minute period as follows: 3 minutes on, 3 minutes off. This is a destructive test. Mechanical or electrical rejects may be used.

15. Tubes meeting the initial test requirements after stability and survival-rate life tests shall be accepted under this specification.

16. Subject tubes to shock in positions X, Y, and Z for a total of 15 blows, five blows in each position.

17. Vibrate in positions X and Y only for a total of 64 hours, 32 hours in each position.

18. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as a part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

19. Use basic test method 2126 (for receiving tubes) for seal strain test of this ceramic tube type.

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User activities:
Navy - AS, OS, MC, CG

Preparing activity:
Navy - EC

Agent:
DLA - ES

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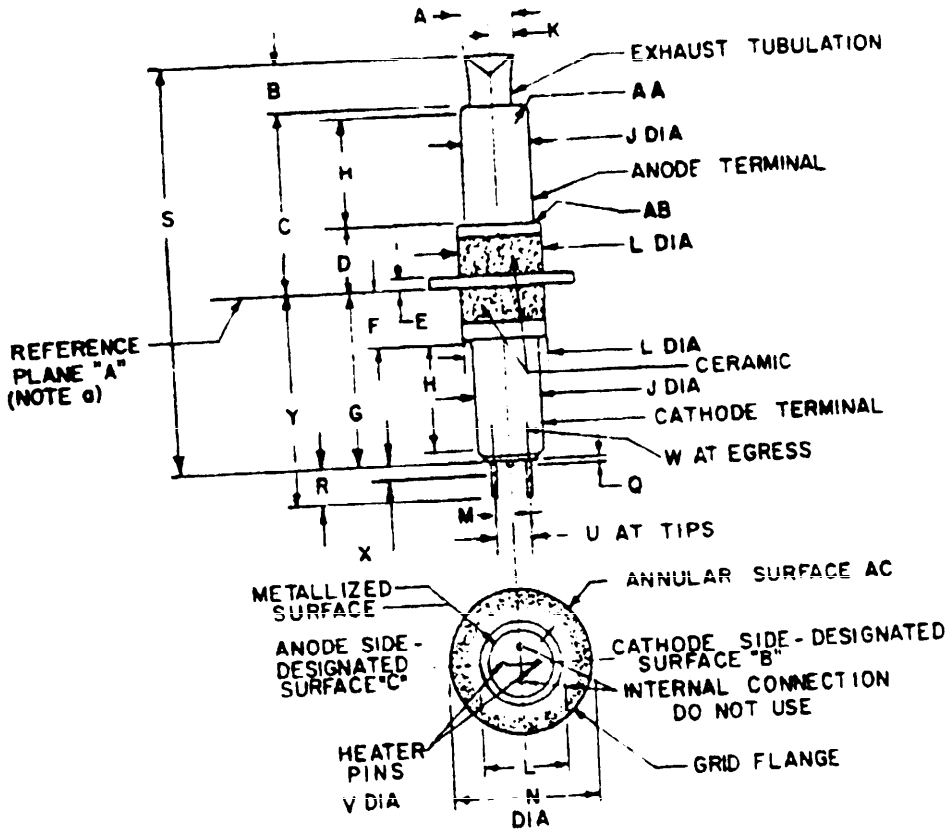


FIGURE 1. Outline drawing of electron tube type 8727.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses		Notes
	Minimum	Maximum	
Quality conformance inspection, part 1			
C	.555 (14.10)	.605 (15.36)	
G	.535 (13.59)	.575 (14.60)	
H	.320 (8.13)		
Y	.650 (16.51)	.700 (17.78)	
Quality conformance inspection, part 2			
A		.230 (5.84)	
B		.180 (4.57)	
D	.165 (4.19)	.205 (5.20)	
E	.049 (1.24)	.055 (1.40)	
F	.120 (3.05)	.150 (3.81)	
J	.245 (6.22)	.252 (6.40)	b & d
K		.115 (2.92)	
L	.335 (8.51)	.355 (9.01)	
M	.048 (1.22)	.068 (1.73)	
N	.547 (13.89)	.557 (14.15)	c
Q		.010 (.25)	
R	.095 (2.41)	.125 (3.17)	
S		1.360 (34.54)	
U	.095 (2.41)	.135 (3.43)	
V	.020 (.51)	.030 (.76)	
W	.110 (2.79)	.120 (3.05)	
X		.065 (1.65)	e
Reference dimensions			
AA	.015 (.38) Rad.		
AB	.030 (.76) Rad.		
AC	.060 (1.52)		

NOTES:

a. Reference plane "A" is defined as that plane against which annular surface "B" of the grid flange abuts.
b. The axis of the anode terminal shall be coincident with the axis of the cathode terminal within .010 inch (.25 mm).
c. The axis of the cathode terminal shall pass through the exact center of the grid flange (dimension N) within .005 inch (.13 mm).
d. With annular surface "B" resting on reference plane "A", the axis of the cathode terminal (dimensions H and J) shall be within 2 degrees of a line perpendicular to reference plane "A".
e. Leads shall be tinned except as specified by dimension X. Untinned length of leads shall be measured from end of ceramic stem wafer.

FIGURE 1. Outline drawing of electron tube type 8727 - Continued.

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